

FINAL PROJECT

Tasks

- **Download/import** the dataset from the given URL.
- Understand the dataset and gain the **domain knowledge**.
- Define the **problem statement(s)** and outline the **project goals**.
- **Import/load** the dataset and necessary libraries/modules.
- Apply **data cleaning** and **preparation**. Please clearly explain what changes you made to the dataset and why.
- Apply **explorative data analysis** techniques to understand the dataset. Please clearly explain each technique you applied and why. Hint: visualization, correlation, mapping, ...
- Apply all the necessary **data pre-processing methods** to transform the data and make it applicable for ML/DL models. Please clearly explain each method you use and why.
- Build multiple **Machine Learning models: at least 1 ensemble, 1 deep learning and at least 5 models in total**.
- **Use advanced techniques to improve the model**. Please clearly explain the reasons for your choice of models, each step you apply in the building process and the results of your models.
- Compare your ML models by using **evaluation techniques/metrics** and explain your evaluation results.
- Apply **tunning methods** to improve the accuracy of your model and compare with the previous models.
- Make **conclusions** from your process and results/evaluation.

IMPORTANT: Please state your name for each part you have contributed.

Submission

Submission Requirements

- **Jupyter Notebook (.ipynb)**: Team code, outputs, answers, and task explanations.
- **Presentation Slides**: Team presentation.

Deadline: Upload all files to Moodle by **21.07.2026, 23:59**.